THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 100 and 1000 (both excluded) is: 1

(a) 20 **(b)** 22 **(c)** 21 **(d)** 23

2. For the expression $n^2 - n + 11$, the smallest natural number n for which the value is NOT prime is: 1

(a) 12 **(b)** 10 **(c)** 13 **(d)** 11

3. For the expression $n^2 + n + 191$, the smallest natural number n for which the value is NOT prime is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

4. The sum of 5 consecutive integers is 180. The largest of them is: 1

(a) 38 **(b)** 40 **(c)** 39 **(d)** 37

5. If n is any integer not divisible by 3, then n^2 leaves a remainder ___ on division by 3. For 7 this gives: 1

(a) 3 **(b)** 1 **(c)** 0 **(d)** 2

6. For the two-digit number 83, the value of (number − digit sum) ÷ 9 is: 1

(a) 8 **(b)** 10 **(c)** 9 **(d)** 7

7. If the sides of a triangle are 9, 40 and x (with x the longest side) and it is right-angled, then x is: 1

(a) 43 **(b)** 42 **(c)** 41 **(d)** 40

8. The smallest natural number with exactly 6 factors (counting 1 and itself) is: 1

(a) 10 **(b)** 13 **(c)** 12 **(d)** 16

9. The number of perfect squares between 250 and 2500 (both excluded) is: 1

(a) 36 **(b)** 34 **(c)** 35 **(d)** 33

10. For the expression $n^2 - n + 17$, the smallest natural number n for which the value is NOT prime is: 1

(a) 17 **(b)** 18 **(c)** 16 **(d)** 19

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Use the converse of the Baudhāyana–Pythagoras theorem: for what value of x is a triangle with sides $20, 21, x$ (x the longest) right-angled? **2**

12. Show that "If a number is divisible by 4 and by 6, then it is divisible by 24" is false. Give a counterexample. **2**

13. Verify that $n^2 - 1$ is a multiple of 8 for the odd number $n = 11$. Write the converse of "If n is odd, then $n^2 - 1$ is a multiple of 8". **2**

14. Use the converse of the Baudhāyana–Pythagoras theorem: for what value of x is a triangle with sides $11, 60, x$ (x the longest) right-angled? **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the sum of the squares of two consecutive integers is odd. Verify it for 8 and 9. **3**

16. Prove: if 3 divides x and x divides y, then 3 divides y. Verify with $x = 12$ and $y = 60$. **3**

17. For $m = 5$ and $n = 2$ show that $(21,\ 20,\ 29)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. **3**

OR

Prove that the sum of the squares of two consecutive integers is odd. Verify it for 12 and 13. **3**

18. Prove: if 7 divides x and x divides y, then 7 divides y. Verify with $x = 14$ and $y = 84$. **3**

OR

For $m = 6$ and $n = 1$ show that $(35,\ 12,\ 37)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the product of three consecutive integers is divisible by 6. Verify it for $5, 6, 7$. **5**

OR

Prove that the square of any odd integer is 1 more than a multiple of 8. Verify it for 9. **5**

20. Prove that the product of three consecutive integers is divisible by 6. Verify it for $9, 10, 11$. **5**

OR

Prove that the square of any odd integer is 1 more than a multiple of 8. Verify it for 13. **5**

END OF QUESTION PAPER — Total: 40 Marks